JAVA CODING FOR KIDS

Learn by doing 10 ready to type game in java

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Setting up for fun.

Learning to code is most fun when you're creating something you can play with! By typing and building these simple games, you're not just learning Java programming concepts—you're also developing problem-solving skills and understanding how code works in real-world applications.

The best way to learn is by doing. As you type each line of code, you'll see how it all comes together to create a working game. You'll learn about variables, loops, and conditional statements in a hands-on way that makes sense. Plus, you can modify the game to make it your own, which is a great way to experiment and learn even more!

So, get started and build your own games. You'll be amazed at how quickly you can create something cool and fun!

How to get started?

If you have already an IDE you can use that or you can type your games online and play without installing extra software.

Here is two simple path you can take to get ready:

1. Online Fun

Open a Web Browser:
Use a browser like Google Chrome, Firefox, or Safari.

Search for "Online Java Compiler":
Type these exact words in the search bar and press Enter.

Choose a Compiler:
Look at the search results and pick a compiler that looks easy to use. Some popular options include:

JDoodle: Known for its simplicity and ease of use.

TutorialsPoint: Offers a comprehensive environment with features like code execution and sharing.

OneCompiler: Provides a robust editor with features like auto-completion and error checking.

2. Use an IDE

If you have the possibility to set up an IDE (Integrated Development Environment) locally with the help of a parent or friend, you should try that! IDEs like IntelliJ IDEA or Eclipse make coding easier and provide powerful tools to help you learn Java.

However, if setting up an IDE feels too complicated right now, don’t worry! You can start coding online using one of the compilers mentioned above. It’s simple, quick, and perfect for beginners.

1 Hello World!

```
import java.util.Scanner;

public class HelloName {
public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);

    System.out.print("What is your name? ");
    String name = scanner.nextLine();

    System.out.println("Hello, " + name + "!");

    scanner.close();
    }
}
```

How Kids Can Modify This Game

1. **Add More Colors:** Add more colors to the colors array to make the game more challenging.
2. **Change the Number of Attempts:** Adjust the number of attempts allowed.
3. **Use Different Themes:** Instead of colors, use animals, fruits, or any other theme that kids enjoy.
4. **Add Hints:** Provide hints if the guess is close (e.g., "Your guess is a similar color").

2 Guessing Games

```
    import java.util.Random;
    import java.util.Scanner;

    public class GuessTheNumber {
    public static void main(String[] args) {
        Random random = new Random();
        int secretNumber = random.nextInt(100) + 1;
        int attempts = 0;
        boolean guessedCorrectly = false;

        Scanner scanner = new Scanner(System.in);
        System.out.println("Welcome to Guess the Number!");
        System.out.println("I'm thinking of a number between 1 and
100.");

        while (!guessedCorrectly) {
            System.out.print("Enter your guess: ");
            int guess = scanner.nextInt();
            attempts++;

            if (guess == secretNumber) {
                guessedCorrectly = true;
            } else if (guess < secretNumber) {
                System.out.println("Too low! Try again.");
            } else {
                System.out.println("Too high! Try again.");
            }
        }

        System.out.println("Congratulations! You guessed the number
in " + attempts + " attempts.");
        scanner.close();
        }
    }
```

How Kids Can Modify This Game

1. **Limit the Number of Attempts – Add a maximum number of guesses (e.g., 10) before the game ends.**
2. **Add a Hint System – Give hints when the guess is very close (within 5).**
3. **Change the Number Range – Reduce the range to make it easier (e.g., 1-20 instead of 1-100).**
4. **Funny Responses – Replace "Too high!" or "Too low!" with fun messages.**
5. **Multiplayer Mode – Let one player choose a number while another guesses.**
6. **Play with Sounds (Using Emojis or ASCII Art) – Add emojis or fun ASCII art when they win.**

```
    import java.util.Random;
    import java.util.Scanner;

    public class GuessMyColor {
        public static void main(String[] args) {
            Random random = new Random();
            String[] colors = {"Red", "Blue", "Green", "Yellow",
"Purple"};
            String secretColor =
colors[random.nextInt(colors.length)];
            int attempts = 0;
            boolean guessedCorrectly = false;

            Scanner scanner = new Scanner(System.in);
            System.out.println("Welcome to Guess My Color!");

            while (!guessedCorrectly && attempts < 5) {
                System.out.print("Guess a color (Red, Blue, Green,
Yellow, Purple): ");
                String guess = scanner.nextLine().trim();
                attempts++;

                if (guess.equalsIgnoreCase(secretColor)) {
                    guessedCorrectly = true;
                } else {
                    System.out.println("Try again!");
                }
            }

            if (guessedCorrectly) {
                System.out.println("Congratulations! You guessed the
color in " + attempts + " attempts.");
            } else {
                System.out.println("Sorry, you didn't guess it. The
color was " + secretColor + ".");
            }
            scanner.close();
        }
    }
```

How Kids Can Modify This Game

7. **Add More Colors:** Add more colors to the colors array to make the game more challenging.
8. **Change the Number of Attempts:** Adjust the number of attempts allowed.
9. **Use Different Themes:** Instead of colors, use animals, fruits, or any other theme that kids enjoy.
10. **Add Hints:** Provide hints if the guess is close (e.g., "Your guess is a similar color").

3 Interactive Games

4 Advance Projects

```
import java.util.Scanner;

public class ConnectFour {
    private static char[][] board = new char[6][7];
    private static char currentPlayer = 'X';

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        initializeBoard();

        while (true) {
            printBoard();
            System.out.println("Player " + currentPlayer + ", choose a column
      (1-7): ");
            int column = scanner.nextInt() - 1;

            if (dropPiece(column)) {
                if (checkWin()) {
                    printBoard();
                    System.out.println("Player " + currentPlayer + " wins!");
                    break;
                }
                currentPlayer = (currentPlayer == 'X') ? 'O' : 'X';
            } else {
                System.out.println("Column is full. Try again.");
            }
        }
        scanner.close();
    }

    private static void initializeBoard() {
        for (int i = 0; i < 6; i++) {
            for (int j = 0; j < 7; j++) {
                board[i][j] = '-';
            }
        }
    }

    private static void printBoard() {
        for (int i = 0; i < 6; i++) {
            for (int j = 0; j < 7; j++) {
                System.out.print(board[i][j] + " ");
            }
            System.out.println();
        }
    }

    private static boolean dropPiece(int column) {
        for (int i = 5; i >= 0; i--) {
            if (board[i][column] == '-') {
                board[i][column] = currentPlayer;
```

```
                return true;
            }
        }
        return false;
    }

    private static boolean checkWin() {
        // Check horizontal locations for win
        for (int i = 0; i < 6; i++) {
            for (int j = 0; j < 4; j++) {
                if (board[i][j] != '-' && board[i][j] == board[i][j + 1] &&
      board[i][j] == board[i][j + 2] && board[i][j] == board[i][j + 3]) {
                    return true;
                }
            }
        }

        // Check vertical locations for win
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 7; j++) {
                if (board[i][j] != '-' && board[i][j] == board[i + 1][j] &&
      board[i][j] == board[i + 2][j] && board[i][j] == board[i + 3][j]) {
                    return true;
                }
            }
        }

        // Check positively sloped diagonals
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 4; j++) {
                if (board[i][j] != '-' && board[i][j] == board[i + 1][j + 1] &&
      board[i][j] == board[i + 2][j + 2] && board[i][j] == board[i + 3][j + 3])
      {
                    return true;
                }
            }
        }

        // Check negatively sloped diagonals
        for (int i = 3; i < 6; i++) {
            for (int j = 0; j < 4; j++) {
                if (board[i][j] != '-' && board[i][j] == board[i - 1][j + 1] &&
      board[i][j] == board[i - 2][j + 2] && board[i][j] == board[i - 3][j + 3])
      {
                    return true;
                }
            }
        }

        return false;
    }
}
```

```
import java.util.Scanner;
import java.util.Random;

public class CatchTheStar {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        Random random = new Random();

        // Game settings - kids can modify these values
        int gameWidth = 10;        // Width of the game area
        int playerPosition = 5;    // Starting position (middle)
        int score = 0;             // Starting score
        int rounds = 15;           // Number of rounds to play
        char playerChar = '^';     // Character representing the player
        char starChar = '*';       // Character representing stars
        char obstacleChar = 'X';   // Character representing obstacles

        System.out.println("=== CATCH THE STAR ===");
        System.out.println("Move left (L) or right (R) to catch stars (*) and
      avoid obstacles (X)");
        System.out.println("Stars = 10 points, Obstacles = -5 points");
        System.out.println("Press Enter to start the game!");
        scanner.nextLine();

        // Main game loop
        for (int round = 1; round <= rounds; round++) {
            // Create a falling object (either star or obstacle)
            int objectPosition = random.nextInt(gameWidth) + 1;
            boolean isStar = random.nextBoolean();
            char fallingObject = isStar ? starChar : obstacleChar;

            // Display the falling object
            System.out.println("\nRound " + round + "/" + rounds + " | Score: "
      + score);
            System.out.println();

            // Show the falling object row
            for (int i = 1; i <= gameWidth; i++) {
                System.out.print(i == objectPosition ? fallingObject : " ");
            }
            System.out.println();

            // Show the player row
            for (int i = 1; i <= gameWidth; i++) {
                System.out.print(i == playerPosition ? playerChar : " ");
            }
            System.out.println();
```

```
            // Get player move
            System.out.print("Move (L/R): ");
            String move = scanner.nextLine().toUpperCase();

            // Process player movement
            if (move.equals("L") && playerPosition > 1) {
                playerPosition--;
            } else if (move.equals("R") && playerPosition <
gameWidth) {
                playerPosition++;
            }

            // Check if player caught the object
            if (playerPosition == objectPosition) {
                if (isStar) {
                    System.out.println("You caught a star! +10 points");
                    score += 10;
                } else {
                    System.out.println("Oh no! You hit an obstacle! -5
points");
                    score -= 5;
                }
            } else if (isStar) {
                System.out.println("You missed a star!");
            } else {
                System.out.println("You avoided an obstacle!");
            }
        }

        // Game over - display final score
        System.out.println("\n=== GAME OVER ===");
        System.out.println("Final score: " + score);

        // Rate the performance
        if (score >= 100) {
            System.out.println("Amazing! You're a star catcher champion!");
        } else if (score >= 50) {
            System.out.println("Great job! You're getting good at this!");
        } else if (score >= 0) {
            System.out.println("Not bad! Keep practicing!");
        } else {
            System.out.println("Better luck next time!");
        }

        scanner.close();
    }
}
```

What kids can easily modify:

- Change the game width to make it easier or harder
- Change the characters used for the player, stars, and obstacles
- Adjust the scoring system (how many points for stars/obstacles)
- Change the number of rounds
- Modify the final messages based on score
- Add new features (like different types of falling objects)

The game uses only basic Java concepts (variables, loops, conditionals, input/output), making it accessible for beginners. Kids can run this in any Java environment, including online IDEs if they don't have Java installed locally.

5 Make it you own!

Other than help you make these games a little customize fro your own fun and taste I want to give you the possibility to take part of a a big challenge: make your own game from scratch! Are you ready?

Try to do you own game and test it. If you are happy with the result you can send it to me for a review. The best game will became part of the next version of this book and if your game is selected you will get the book for free!

Here are some tipp to try you luck.

1. The easy way: try use code elementes or pieces of code you found in the games you already build. If you know these code “pieces” already worked it will be easy to make your game changing the logic and piecing together different part from differt game is not illigal. Your fantasy is the limit.
2. The Hard way: start from scratch if you are brave enogh and give it a try. Here the sky is the limit.

In Both cases remember do not make it to long, the goal is to keep it fun to type in and help other kids like you learn!

As well as helping you tweak these games a little for your own enjoyment and taste, I want to give you the opportunity to take part in a great challenge: to create your own game from scratch! Are you ready?

Try making and testing your own game. If you are happy with the result, send it to me for a review. The best game will become part of the next version of this book, and if your game is selected, you will get the book for free!

Here are some tips to try your luck.

1. The easy way: try to use code elements or pieces of code that you have found in the games you have already made. If you know these code "pieces" already worked, it will be easy to make your game, changing the logic and piecing together different parts from differt game is not illegal. Your imagination is the limit.

2. The hard way: start from scratch, if you are brave enough to try. The sky is the limit.

In both cases, remember not to take too long, the goal is to keep it fun and to help other kids like you learn!

Here you can send your game and question:

marcogara24@gmail.com

Not only can you tweak these games to match your own style and creativity, but I also have an exciting challenge for you: **Create your own game from scratch!** Are you ready?

Try making and testing your very own game. If you're happy with the result, send it to me for a review! The best game will be featured in the next version of this book, and if your game is selected, **you'll get the book for free!**

Here are some tips to help you on your journey:

1. **The Easy Way** – Use code elements or pieces of code from the games you've already made. Since you know these parts work, you can mix and match them to create something new. Think of it like building with LEGO—your imagination is the only limit!
2. **The Hard Way** – Start completely from scratch, if you're feeling brave! This way, you can build something totally unique. The sky's the limit!

In both cases, **keep your code short and simple!** The goal is to make it easy to understand, fun to play, and helpful for other kids who want to learn.

Formatting style (not part of the book)

-code snippet Page 5, 9-12 its ok

14-15-16 is the same page needs to find the best or a mix of them